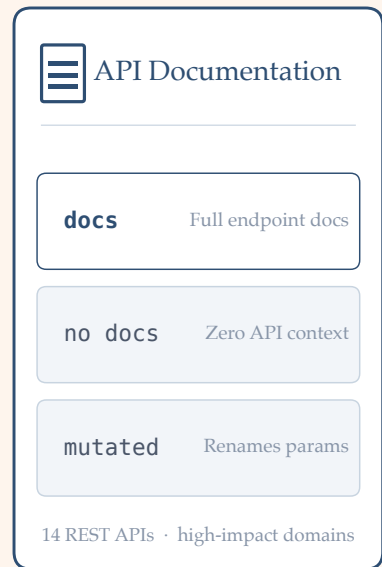


SYNTHESIS (offline, per API)



INSTRUCTIONS.md

server spec · key endpoints



Tool-Maker LLM

generates MCP server code

via file-system tools

read · list · glob

grep · write · edit

EVALUATION (online, per task)



MCP Server

N tool functions

@mcp.tool()

API client

auth + HTTP

HTTP resp ↑ ↓ HTTP req



Live API

or sandbox when available

Test task

tasks.yaml · M per API



Agent LLM

uses MCP tool calls

trajectory ≤ 15 turns

results

tool calls



Judge LLM

evaluates tool usage

and task completion



Task Verdict

PASS

agent completed task correctly

FAIL

tool-side failure (see cause)

UNDEF

agent / env / task confound

Failure cause taxonomy

TOOL_COVERAGE

TOOL_SCHEMA

TOOL_IMPL.

TOOL_DOCS

Server quality (indep. of agent)

SERVER_SUFFICIENT

did the server provide a correct, usable tool?

YES / NO / UNKNOWN